Force/Torque Sensor-Free Online Payload Estimation for a Pneumatically Driven Robot

Kathrin Hoffmann[1], Christian Trapp[2], Alexander Hildebrandt[2], and Oliver Sawodny[1]

***Abstract*— Robots with pneumatic drives have advantageous physical properties and are lightweight and cost-effective. During operation, a pneumatic robot handles objects of different payloads, which are often not known in advance. Determining the mass of this load is crucial for user assistance functions, trajectory generation and model-based control to increase both productivity and ease of use. In this work, a model-based approach for estimating the mass of a payload gripped by a pneumatically driven robot is presented. This approach does not require additional force or torque sensors, but uses the joint angle and pressure sensors already installed for control purposes. From these signals, the non-measurable effective driving torque is calculated model-based, taking into account the particularities of the pneumatic drives. As a basis for the estimation, the robot equations of motion are formulated such that they are linear in the dynamic parameters of the robot. An optimization problem is set up and solved with a recursive least squares method to estimate the payload online. This is extended by a logic to activate the estimation when a payload is picked. The algorithm is validated experimentally on a pneumatic robot performing a typical pick-and-place task.**

I. INTRODUCTION

In robotics, pneumatic actuators are an interesting alternative to electric actuators because of their physical properties. Pneumatic drives are inherently compliant due to the compressibility of air. Moreover, they are direct drives without gears, which does not only make them light in weight, but also leads to a small collision torque, since little inertia needs to be decelerated. These are advantageous safety features, so that pneumatically driven robots are well suited for human-robot collaboration.

Today's industrial robots perform increasingly complex tasks in which the mass of the load they transport is relevant for several reasons. First, payload estimation can assist the user by providing the mass information, or to verify or add data to object recognition. Second, the maximum robot velocity and its derivatives are physically constrained and depend on the payload. Knowing the payload enables dynamic trajectory planning to reduce the cycle time and increase productivity. Third, many advanced robot control algorithms are based on a model of the robot dynamics. This requires the robot dynamic parameters, which are the mass, the coordinates of the center of gravity, and the elements of the inertia tensor of each robot segment. In the present pneumatic lightweight robot with low inertia and stiffness, the influence of the load on the robot dynamics is larger than for a comparable electric robot. Therefore, the load should be known approximately for model-based control.

Approaches to the model-based estimation of robot dynamic parameters in general, and load parameters in particular are described in the literature, such as [1], [2], [3], [4], [5], [6], [7]. Some of these works aim to identify as many dynamic parameters of the robot links as possible. If a validated dynamic model of the robot without payload is already available, its parameters are taken as given and only the payload parameters need to be identified. For the present robot, the kinematic and dynamic parameters of each link are available from CAD data with sufficient accuracy, which leaves the unknown payload parameters to be estimated.

For the estimation of dynamic parameters, the knowledge of torques and/or forces is required. Therefore, in most of the literature, there are either torque sensors installed in the joints or a force sensor is installed at the endeffector. In order to reduce cost and complexity, it is desirable to implement payload estimation without such additional sensors. Instead, the joint torques can be computed from an actuator model in combination with sensor signals already measured in the actuator for control purposes. In the literature, where payload estimation is usually implemented on robots with electric drives, this is rarely done, since it implies that losses and dynamic effects in the motors and transmissions need to be identified [3]. The latter introduces additional model uncertainty. For pneumatic drives, which are direct drives without gears, these effects do not occur. Instead, other properties of the pneumatic drives need to be taken into account and modeled suitably, as discussed below. This is done in this research, so that the payload estimation works without additional sensors, just from the existing measurement of actuator pressures and joint angles along with an actuator model.

As an alternative to these model-based approaches, also characteristic diagrams can be recorded, or nonparametric models can be learned based on data [8]. These require extensive training in advance, whereas the proposed model-based approach requires only a state-of-the-art CAD model and a priori actuator friction identification of the pneumatic robot. The friction identification is necessary because in the present case, only the driving torque from the measured pressures in the actuator can be determined, but there is no measurement of the effective driving torque after subtracting friction. This means that there is an input side torque measurement, but no output side torque measurement, since no extra sensor is installed. This setup follows directly from

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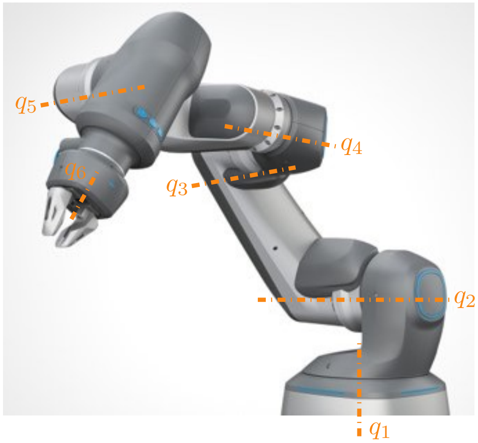

Fig. 1. Pneumatic robot and rotation axes of the joints.

the pneumatic drive technology, in which pressure sensors are installed for control purposes, while at the same time there is significant friction in the seals around pressurized components. These particularities of pneumatic drives are taken into account in this work. Consequently, the approach from this work is applicable to pneumatic manipulators with arbitrary open-chain kinematics, requiring no extra sensors, but using existing pressure sensors.

Since the parameters to be estimated need to be sufficiently excited during the measurement, there is a vast amount of literature on how to generate exciting trajectories [9], [10]. In practice, however, it is often necessary to execute a robot application prespecified by the user, without the possibility to modify the trajectory, e.g. due to no access to the trajectory generation software. For this reason, the goal of this work is to estimate the payload during an existing real-world robot application on the basis of a given trajectory for a typical pick-and-place cycle. The estimation is carried out online.

This paper is organized as follows: The dynamic model of the pneumatically driven robot is derived in Section II such that it is linear in the dynamic parameters. In Section III, the payload estimation approach is introduced. Experimental results to validate this approach are presented and discussed in Section IV for a typical pick-and-place application with different payloads. In Section V, this paper is summarized.

II. Robot Model for Parameter Estimation

In this section, the dynamic model of the pneumatically driven robot is introduced and reformulated in regressor form, suitable for the subsequent parameter estimation. The robot is an open-chain manipulator with six revolute joints and a workspace radius of $650\,\mathrm{mm}$. Their rotation axes are labeled in Fig. 1, and $\boldsymbol{q} \in \mathbb{R}^{n_{\mathrm{DOF}}}$ with $n_{\mathrm{DOF}} = 6$ is the vector of joint angles.

The robot is modeled as a rigid body system and its equations of motion are derived with the Lagrange formalism based on geometric and dynamic parameters (for each link the mass, center of gravity, and inertia) from CAD data. This yields the equation of motion

$$\boldsymbol{M}(\boldsymbol{q})\,\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\,\dot{\boldsymbol{q}} + \boldsymbol{G}(\boldsymbol{q}) = \boldsymbol{\tau}(\boldsymbol{p}) - \boldsymbol{\tau}_{\mathrm{f}}(\dot{\boldsymbol{q}})\,, \tag{1}$$

where $\dot{\boldsymbol{q}}$ and $\ddot{\boldsymbol{q}}$ denote the joint velocities and accelerations, $\boldsymbol{M}(\boldsymbol{q})$ is the inertia matrix, $\boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\,\dot{\boldsymbol{q}}$ the vector of Coriolis and centrifugal terms, and $\boldsymbol{G}(\boldsymbol{q})$ the vector of gravitational torques. The right-hand side is specific to the pneumatic drives with pressure-dependent driving torque $\boldsymbol{\tau}(\boldsymbol{p})$ and friction torque $\boldsymbol{\tau}_{\mathrm{f}}(\dot{\boldsymbol{q}})$.

The six pneumatic actuators are rotary drives. Each consists of two pressure chambers separated by a rotatable swivel wing and a static chamber seal. In each chamber, the pressure is controlled individually. A pressure difference generates the driving torque of drive i

$$\tau_i(\boldsymbol{p}) = A_i r_{\mathrm{m},i}(p_{1,i} - p_{2,i}) \tag{2}$$

with A_i being the effective area and $r_{\mathrm{m},i}$ the effective radius of the swivel wing. With these drives being built into the robot, for each robot joint, the measured signals are the angle q_i and the chamber pressures $p_{1,i}$ and $p_{2,i}$. To avoid leakage, the pressure chambers and the rotatable swivel wing have to be sealed all around. As a result, there occurs friction in the seals, up to $10\,\%$ of the maximum driving torque. This is modeled as the sum of Coulomb and viscous friction

$$\tau_{\mathrm{f},i} = f_{\mathrm{c},i}\tanh\left(\frac{\dot{q}_i}{\dot{q}_{\mathrm{c},i}}\right) + f_{\mathrm{v},i}\dot{q}_i, \tag{3}$$

where $f_{\mathrm{c},i}$ is the Coulomb friction torque and $f_{\mathrm{v},i}$ is the viscous friction coefficient, which are identified a priori. There is no Stribeck effect, because a suitable lubricant is used. Overall, the right-hand side of (1) describes the effective driving torque, without requiring a torque sensor, and is specific to the pneumatic drive technology.

A. Dynamic model in regressor form

The left-hand side of (1) is a function of the geometric and dynamic parameters of the robot. Notably the dynamic model is linear in the robot dynamic parameters [1], [12]. These parameters are for each link i

$$\begin{aligned}\boldsymbol{\theta}_i = \big[&m_i \quad m_i r_{\mathrm{cog},x,i} \quad m_i r_{\mathrm{cog},y,i} \quad m_i r_{\mathrm{cog},z,i} \\ &I_{xx,i} \quad I_{xy,i} \quad I_{xz,i} \quad I_{yy,i} \quad I_{yz,i} \quad I_{zz,i}\big]^{\mathsf{T}} \in \mathbb{R}^{10},\end{aligned} \tag{4}$$

i.e., the mass m_i, the coordinates of the center of gravity $\boldsymbol{r}_{\mathrm{cog},i}$, occurring as the first moment of inertia in a product with the mass, and the unique components of the inertia tensor $\boldsymbol{I}_i$. In the above nomenclature, link i is the one located at the output side of joint i. For the entire six-link robot the so-called standard inertial parameters are

$$\boldsymbol{\theta} = [\boldsymbol{\theta}_1^{\mathsf{T}}\ \boldsymbol{\theta}_2^{\mathsf{T}}\ \cdots\ \boldsymbol{\theta}_6^{\mathsf{T}}]^{\mathsf{T}} \in \mathbb{R}^{10 n_{\mathrm{DOF}}}. \tag{5}$$

With this, (1) can be rewritten as

$$\boldsymbol{Y}(\boldsymbol{q},\dot{\boldsymbol{q}},\ddot{\boldsymbol{q}})\,\boldsymbol{\theta} = \boldsymbol{\tau}\left(\boldsymbol{p}\right) - \boldsymbol{\tau}_{\mathrm{f}}\left(\dot{\boldsymbol{q}}\right), \tag{6}$$

where $\boldsymbol{Y} \in \mathbb{R}^{n_{\mathrm{DOF}} \times 10 n_{\mathrm{DOF}}}$ is the regressor matrix, which is a function of the robot geometry. Starting from the existing derivation of (1) by means of the Lagrange formalism, also the regressor form (6) is derived with this method. An explicit Lagrangian formulation of the regressor is obtained from the procedure described in [13]. It is computed offline and evaluated online for measurements of $\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}}$.

B. Model formulation for payload estimation

The robot endeffector, which picks up the payload, is firmly connected to the last link of the robot. Consequently, a payload enters into the equations of motion in the same way as the dynamic parameters of link 6. The full payload dynamic parameter vector is

$$\begin{aligned}\boldsymbol{\theta}_{\text{payload}} = \big[&m \quad mr_{\text{cog},x} \quad mr_{\text{cog},y} \quad mr_{\text{cog},z} \\ &I_{xx} \quad I_{xy} \quad I_{xz} \quad I_{yy} \quad I_{yz} \quad I_{zz}\big]^{\mathsf{T}} \in \mathbb{R}^{10},\end{aligned} \tag{7}$$

where $\boldsymbol{r}_{\text{cog}}$ and $\boldsymbol{I}$ are expressed with respect to the body-fixed coordinate frame of link 6. The left-hand side of (6) is composed as

$$\begin{bmatrix}\boldsymbol{Y}_1 & \boldsymbol{Y}_2 & \cdots & \boldsymbol{Y}_6\end{bmatrix}\begin{bmatrix}\boldsymbol{\theta}_1 \\ \boldsymbol{\theta}_2 \\ \vdots \\ \boldsymbol{\theta}_6 + \boldsymbol{\theta}_{\text{payload}}\end{bmatrix} \tag{8}$$

in the presence of a payload. Therein, $\boldsymbol{Y}_6 \in \mathbb{R}^{n_{\text{DOF}} \times 10}$ are the columns of $\boldsymbol{Y}$ corresponding to $\boldsymbol{\theta}_6$. A payload causes joint torques $\boldsymbol{\tau}_{\text{payload}} = \boldsymbol{Y}_6\boldsymbol{\theta}_{\text{payload}}$. The overall joint torques are regrouped as

$$\underbrace{\boldsymbol{Y}\boldsymbol{\theta}_{\text{robot}}}_{\boldsymbol{\tau}_{\text{robot}}} + \underbrace{\boldsymbol{Y}_6\boldsymbol{\theta}_{\text{payload}}}_{\boldsymbol{\tau}_{\text{payload}}} = \boldsymbol{\tau}\left(\boldsymbol{p}\right) - \boldsymbol{\tau}_{\text{f}}\left(\dot{\boldsymbol{q}}\right), \tag{9}$$

being the sum of the torques $\boldsymbol{\tau}_{\text{robot}}$ from the robot without payload, and the torques $\boldsymbol{\tau}_{\text{payload}}$ from the payload. Therein, the following terms are calculated model-based at each measurement point in time. The driving torque $\boldsymbol{\tau}(\boldsymbol{p})$ results from the measured pressures using (2). The friction torque $\boldsymbol{\tau}_{\text{f}}(\dot{\boldsymbol{q}})$ is approximated by the model (3). Note that the right-hand side of (9) is subject to friction torque uncertainties and that no measurement of the effective drive torque after subtracting friction is available. The standard inertial parameters $\boldsymbol{\theta}$ are extracted from the robot CAD model and the full regressor $\boldsymbol{Y}$ is calculated incorporating the robot geometry from the same source. Further, $\boldsymbol{Y}$ depends on $\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}}$, of which $\boldsymbol{q}$ is available from measurement, and its derivatives are calculated with a Savitzky-Golay filter [14]. In total, only $\boldsymbol{\theta}_{\text{payload}}$ is unknown and is to be estimated.

III. Payload estimation

Rearranging the model (9) from the previous section yields the linear equation system

$$\underbrace{\boldsymbol{\tau}\left(\boldsymbol{p}\right) - \boldsymbol{\tau}_{\text{f}}\left(\dot{\boldsymbol{q}}\right) - \boldsymbol{\tau}_{\text{robot}}}_{\boldsymbol{\tau}_{\text{fit}} \in \mathbb{R}^{n_{\text{DOF}}}} = \boldsymbol{Y}_6\boldsymbol{\theta}_{\text{payload}}. \tag{10}$$

Simulations of typical robot motions with payloads according to the robot specifications indicate that the first moment of inertia $m\boldsymbol{r}_{\text{cog}}$ has a much larger influence on the joint torques than the moment of inertia tensor $\boldsymbol{I}$. Therefore, the parameters $I_{xx}, I_{xy}, I_{xz}, I_{yy}, I_{yz}, I_{zz}$ of the payload are not considered in the following, but instead a reduced payload parameter vector

$$\tilde{\boldsymbol{\theta}}_{\text{payload}} = \begin{bmatrix}m & mr_{\text{cog},x} & mr_{\text{cog},y} & mr_{\text{cog},z}\end{bmatrix}^{\mathsf{T}} \tag{11}$$

and corresponding reduced regressor $\tilde{\boldsymbol{Y}}_6$ are considered. Generally, it may be the case that only some of the ten payload parameters or linear combinations of them can be identified. The identifiability depends on the manipulator structure and the motion. This is reflected by the fact that in $\boldsymbol{Y}_6$ there can occur zero-filled or linearly dependent columns. This determines the rank of $\boldsymbol{Y}_6$ and consequently the solvability of the linear equation system (10) for the parameters $\boldsymbol{\theta}_{\text{payload}}$. Visually speaking, a parameter is identifiable if it has a unique influence on the joint torque, i.e., the corresponding column in the regressor is nonzero and linearly independent. There exist analytic and numeric methods to determine the minimum identifiable set of inertial parameters of a robot.

Another aspect to be considered when formulating the estimation problem is the number of samples to be used in each estimation step. A first variant is to take just the one measurement from the current time step. This has the advantage that the equation system (10) to be solved for $\boldsymbol{\theta}_{\textbf{payload}}$ is small and that payload changes can be detected the fastest. A disadvantage of this approach is its sensitivity to friction model uncertainties, as they occur with the present robot, so that deviations from the friction model (3) are erroneously estimated into the payload dynamic parameters. A second variant is to include multiple samples up to the current time step. This can be implemented with or without a decreasing weight of older measurements, depending on whether a payload change is to be expected or not, e.g., because of gripping. This increases robustness and also allows for estimating more parameters than the first variant, because some of them may be identifiable over the course of the trajectory but not at each time step. In this case, N measurements from a starting time step $t_{k_{\text{s}}}$ up to the current time step $t_{k_{\text{s}}+N}$ are collected into

$$\underbrace{\begin{bmatrix}\boldsymbol{\tau}_{\text{fit},k_{\text{s}}} \\ \boldsymbol{\tau}_{\text{fit},k_{\text{s}}+1} \\ \vdots \\ \boldsymbol{\tau}_{\text{fit},k_{\text{s}}+N}\end{bmatrix}}_{\bar{\boldsymbol{\tau}}_{\text{fit}}} = \underbrace{\begin{bmatrix}\tilde{\boldsymbol{Y}}_{6,k_{\text{s}}} \\ \tilde{\boldsymbol{Y}}_{6,k_{\text{s}}+1} \\ \vdots \\ \tilde{\boldsymbol{Y}}_{6,k_{\text{s}}+N}\end{bmatrix}}_{\bar{\tilde{\boldsymbol{Y}}}_6}\tilde{\boldsymbol{\theta}}_{\text{payload}}. \tag{12}$$

This is a larger system of equations than (10), but still solvable during robot operation by a suitable method and the computing power in the present robot control system.

A. Optimization problem

With multiple measurements being collected into (12), the equation system is overdetermined. Therefore, it is solved for the parameter vector with a least squares approach

$$\min_{\hat{\tilde{\boldsymbol{\theta}}}_{\text{payload}}} \tfrac{1}{2}\left(\bar{\boldsymbol{\tau}}_{\text{fit}} - \bar{\tilde{\boldsymbol{Y}}}_6\hat{\tilde{\boldsymbol{\theta}}}_{\text{payload}}\right)^{\mathsf{T}}\left(\bar{\boldsymbol{\tau}}_{\text{fit}} - \bar{\tilde{\boldsymbol{Y}}}_6\hat{\tilde{\boldsymbol{\theta}}}_{\text{payload}}\right) \tag{13}$$

to minimize the squared error between the measured and modeled joint torques caused by the payload. The analytic solution to this least squares problem is

$$\hat{\tilde{\boldsymbol{\theta}}}_{\text{payload}} = \left(\bar{\tilde{\boldsymbol{Y}}}_6^{\mathsf{T}}\bar{\tilde{\boldsymbol{Y}}}_6\right)^{-1}\bar{\tilde{\boldsymbol{Y}}}_6^{\mathsf{T}}\bar{\boldsymbol{\tau}}_{\text{fit}}, \tag{14}$$

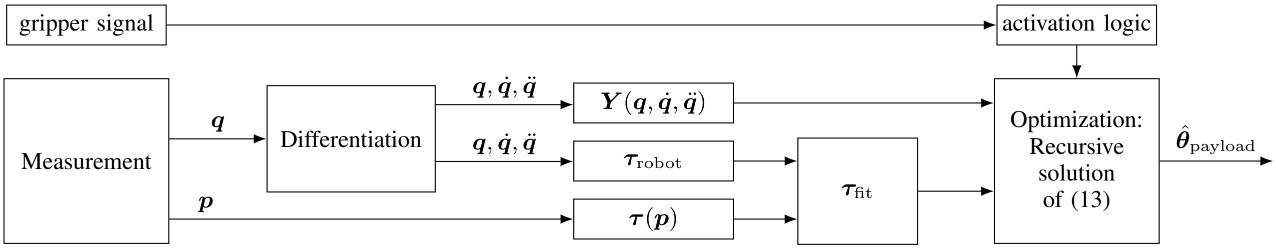

Fig. 2. Structure of the algorithm for online estimation.

which enables an offline parameter identification. To ensure that the estimation problem is well-conditioned, unidentifiable parameters and the corresponding regressor columns should be removed beforehand. In the present case this is done numerically with a QR decomposition of the regressor matrix, see [15].

In contrast to that, for an online estimation, the optimization problem (13) is solved in each time step k with a recursive least squares algorithm [16]. Both approaches are implemented on the experimental setup. The overall flow chart of the payload estimation is depicted in Fig. 2.

B. Application-specific modifications to the estimation problem

Depending on the application, it can be the case that only some of the parameters m, $mr_{\text{cog},x}$, $mr_{\text{cog},y}$ and $mr_{\text{cog},z}$ can be estimated reliably or be estimated at all. With the present robot, in experiments it is possible with suitably exciting trajectories to identify all four of the reduced payload parameters (11). In the remainder of this paper, the estimation algorithm is tested on a typical pick-and-place application. In the experimental setup depicted in Fig. 3, the robot picks a payload at the pick position and drops it into the box at the place position. All payloads are gripped centrically to avoid backlash in the gripper. Consequently, the parameters $r_{\text{cog},x}$ and $r_{\text{cog},y}$ are zero. Moreover, the gripper points downwards during the whole trajectory, so that the parameter $r_{\text{cog},z}$ is barely excited. Therefore, although all four parameters are estimated, only the load mass is considered further.

For the present pick-and-place scenario, the estimation should be active while a payload is gripped, which can only be the case when the gripper, whose control signal is available, is closed. As the payload is placed on a base for pickup, it is still supported by the base at the beginning, when the gripper is closed already, but the payload is not lifted up yet, so that it does not have an effect on the robot yet. Instead, the payload is effective when lifted, i.e., when the z-coordinate of the robot tool center point is above the gripping position. In order to evaluate this condition, the tool center point coordinates are calculated by robot forward kinematics. Fig. 4 shows the Cartesian coordinates of the tool center point during the pick-and-place motion. The time interval of a lifted payload is marked in yellow, as determined from the z-coordinate. If, beyond that, backlash is known to occur between the gripper and the payload, the activation of the estimation has to be parametrized differently to start at an even later point in time.

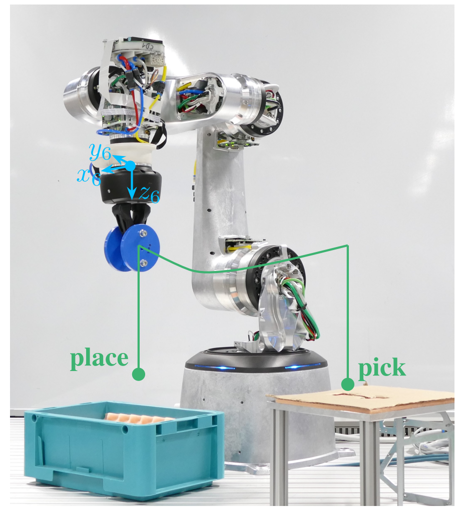

Fig. 3. Experimental setup and pick-and-place trajectory —.

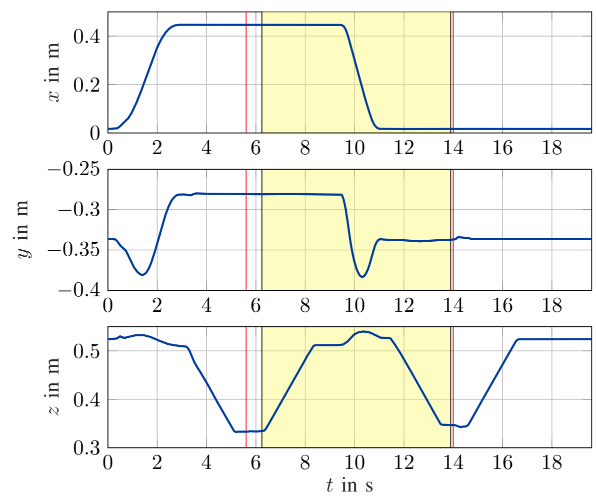

Fig. 4. Coordinates of the tool center point —, closing and opening time of the gripper —, time interval for payload estimation □.

IV. EXPERIMENTAL RESULTS

In the experiments, the robot is controlled via a dSPACE rapid prototyping system with the controllers and measurement executed at 1 kHz. Details on the control algorithm are given in [17]. Since this controller is model-based, a median robot model with a payload of 1 kg is implemented in the controller. It, as well as the controller parametrization, remains the same for all scenarios. Torque errors because of a deviation from this median payload are compensated by the feedback path and disturbance observer in the controller. This resulting overall control torque $\tau(p)$ is the input to the present estimation. For the payload estimation algorithm, the signals are downsampled to 200 Hz, and even smaller sampling frequencies are possible. The robot performs the pick-and-place application described in Sec. III-B with different payloads.

A. Model validation

In order to validate the robot dynamic model (1), the pick-and-place motion is carried out without a payload first. Fig. 5 shows the joint torques $\boldsymbol{\tau}(\boldsymbol{p})$ calculated from the measured pressures along with the robot model. There is a good match between the model and the measurement. The differences between the two that occur result mainly from friction, which is known to be challenging in pneumatic robots. In the present robot model, the friction torque $\boldsymbol{\tau}_{\mathrm{f}}$ in the robot joints is approximated by the static friction model (3). Measurements on the robot joints, however, show that hysteresis occurs. Such an effect can be modeled by dynamic friction models from literature, in which friction is subject to an additional ordinary differential equation. However, a fit of the joint friction measurements leads to a stiff differential equation, which must numerically be solved with step sizes of less than 1 ms. This model is therefore not real-time capable on the experimental setup, which would also make the online payload estimation described here no longer possible. The consequences of the hysteresis are for example visible in the torque τ_1 of the first joint, between 3 and 9 seconds, and from second 12 to the end, where the joint is at rest and no torques other than friction are present. From the static friction model, $\tau_{\mathrm{f},1} = 0\,\mathrm{N\,m}$ is calculated, but the measurement shows a nonzero torque. The torque deviations of each joint are in the range of Coulomb friction, which is $\boldsymbol{f}_{\mathrm{c}} = [1.2, 1.7, 1.2, 0.89, 0.89, 0.37]\,\mathrm{N\,m}$.

B. Estimation of different payloads

In another measurement, the pick-and-place application is executed with picking a payload of 1.027 kg. The corresponding joint torques are plotted in Fig. 5 in dark blue. The biggest torque differences to the measurement without payload occur in joints 2 and 3, on which the gravitational effects from the payload have the biggest influence due to the robot configuration. To this measurement with a 1.027 kg payload, the payload estimation is applied. The estimation is active while the payload is lifted up, as described in Sec. III-B and marked in Fig. 4 and Fig. 5 in yellow.

First, **offline estimation** is tested. To this end, all samples are combined according to (12) and the analytic least squares solution (14) is calculated. This one-time evaluation for the entire interval during which the payload is picked yields a load mass of 0.994 kg. The same experiment is conducted with different payloads and the estimation results are given in Tab. I. These results show that the payload estimation works accurately in a realistic application.

Given the insights on the frictional effects from the previous subsection, one may infer that the accuracy of the estimation increases when times of low joint velocities, where torque uncertainty due to hysteresis is larger, are removed. However, experiments show that this is not the case. Not only friction, but also the inertial effects of the payload are particular distinct when starting and braking a joint motion, providing a good parameter excitation. Therefore, for the estimation all samples are utilized. Beyond that, the payload has the biggest influence on the torques of joints 2 and 3, which is reflected by large coefficients in the regressor. Accordingly, in the regression the measurements of τ_2 and τ_3 are inherently weighted more heavily. The friction-induced deviations, which occur mainly in joints 1 and 4, have little influence on the estimation, since the corresponding joint torques are weighted low. Removing τ_1 and τ_4 a priori from the estimation does not yield a further improvement.

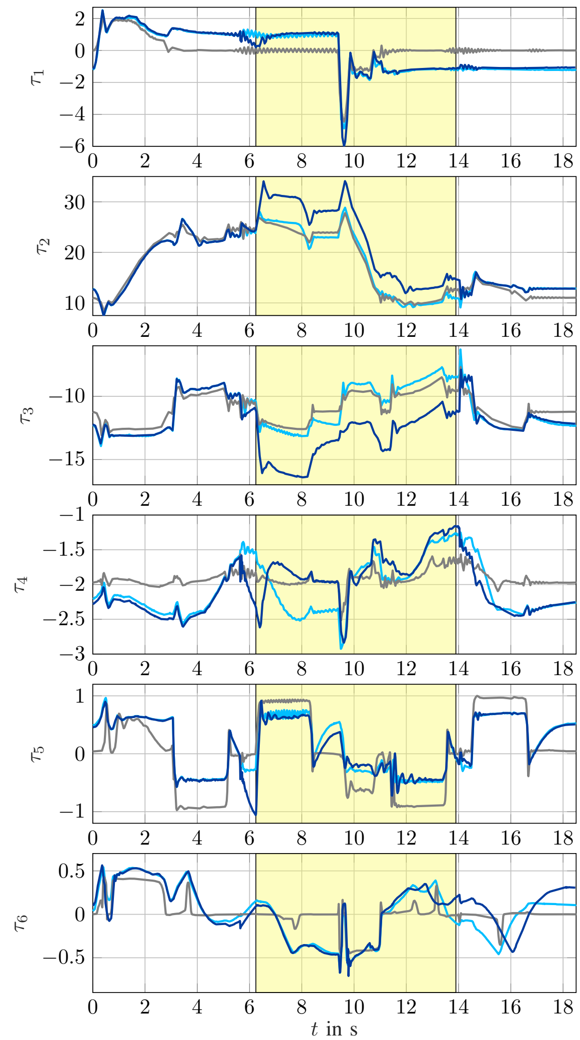

Fig. 5. Joint torques $\boldsymbol{\tau}(\boldsymbol{p})$ from measurement without payload (—), from model $\boldsymbol{M}(\boldsymbol{q})\,\ddot{\boldsymbol{q}}+\boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\,\dot{\boldsymbol{q}}+\boldsymbol{G}(\boldsymbol{q})+\boldsymbol{\tau}_{\mathrm{f}}(\dot{\boldsymbol{q}})$ without payload (—), and from measurment with picking a 1.027 kg payload (—) during the highlighted interval.

TABLE I

ESTIMATED MASSES, STANDARD DEVIATION FROM 20 RUNS, AND CORRESPONDING RELATIVE ERRORS.

real mass in kg	est. mass in kg	standard dev. in kg	relative error
0.245	0.251	0.009	2.5 %
0.445	0.437	0.011	1.8 %
0.650	0.630	0.011	3.1 %
1.027	0.994	0.013	3.2 %
2.055	1.996	0.011	2.9 %

Next, **online estimation** with a recursive least squares approach to solving (13) up to the current sample is tested. It yields the estimated mass depicted in Fig. 6 in blue over time, starting from pickup. The algorithm is initialized with an estimated mass of $0\,\mathrm{kg}$. If more information about the load to be handled is available, other initial conditions are possible. From there, the estimated mass increases strongly at the beginning. Already at the end of the payload lifting motion, the estimate is accurate to $100\,\mathrm{grams}$. With more samples being available for the estimation, it settles around the actual mass. At what point the estimated payload is considered to be sufficiently accurate, and thus valid, depends on the intended use case of the payload estimate. To this end, further uncertainty analysis can be carried out.

To facilitate the understanding of the estimated mass over time, the robot model for the joint torques is simulated with a payload of $1.027\,\mathrm{kg}$, which is not included in Fig. 5 for visibility reasons. When the absolute value of the measured torque is larger than in the simulated model with $1.027\,\mathrm{kg}$ payload, especially due to increased friction, a mass larger than the actual one is estimated, and vice versa. This confirms the physical intuition and explains the curve in Fig. 6 over time. The figure also shows the mass from the offline estimation, as described before. It is equivalent to the online estimate's endpoint, since recursive least squares without forgetting factor was applied in the online case.

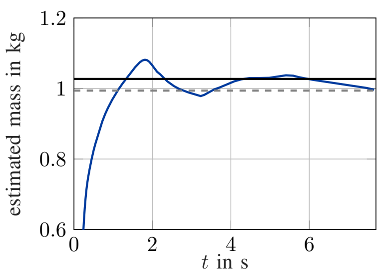

Fig. 6. Payload mass estimated recursively online (—), estimated over entire measurement (– –), and real (—). Estimation and time axis starting at payload pickup.

V. Conclusion

In this paper, an algorithm for estimating the mass of a payload gripped by a pneumatically driven robot without additional force or torque sensors was presented. In the pneumatic robot, the joint torques can be determined from the pressures, measured with existing sensors, and an actuator model. Due to linearity, the joint torques resulting from the payload are calculated based on the dynamic robot model as the difference between the joint torques from the actuator model and the calculated joint torques of the robot without load. A limitation to the estimation accuracy are frictional effects, which cannot be represented exactly in the present real time capable dynamic model, and are a challenge in pneumatic drives. The underlying dynamic robot model was formulated such that it is linear in the payload parameters. A least squares optimization problem was posed to minimize the squared error between these modeled joint torques and the ones calculated from the actuators. Both an analytic offline and a recursive least squares online solution were introduced. This formulation was then adapted according to a real-world application, and the estimation is activated by a logic when a payload is picked up. The algorithm was validated experimentally on a pneumatic robot performing a typical pick-and-place cycle and yields accurate results for different payloads.

Future work includes testing the payload estimation on further robot applications that involve payload handling. According to these scenarios, also the systematics to determine in which phases the estimation is active can be developed further. If an online feedback of the estimated parameters into the model-based controller is desired, a robustness analysis to model uncertainties is recommended.

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